

GEOGRAPHY

PHYSICAL GEOGRAPHY



GEOMORPHOLOGY

MODULE - 22

VOLCANOES

- EXTRUSIVE LANDFORMS
- HOT SPRINGS AND GEYSER

EXTRUSIVE LANDFORMS

1 COMPOSITE VOLCANOES/ STRATOVOLCANOES



- These Volcanoes are characterized by eruption of more viscous lava than basalt.
- These Volcanoes often result in explosive eruptions.
- Along with lava large quantities of pyroclastic material and ashes find their way to the ground
- The materials get accumulated in the vicinity of the vent leading to the formation of layers and this makes the mount appear as composite volcanoes.

EXTRUSIVE LANDFORMS

2

CALDERAS



- Highly explosive volcanoes upon eruption collapse on themselves rather than building any tall structure.
- The collapsed depressions are called Calderas.
- Their explosiveness indicates that magma chamber supplying the lava is not only huge but is also close in vicinity.

EXTRUSIVE LANDFORMS

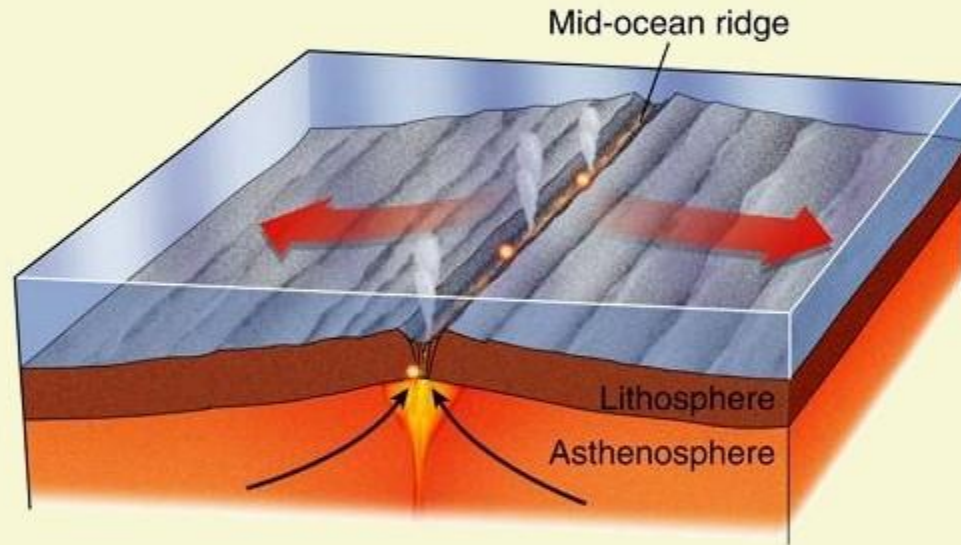
3 FLOOD BASALT PROVINCES



- These volcanoes **outpour highly fluid lava** that flows for long distances.
- There can be a series of flows with some flows attaining thickness of more than 50 metres.
- The **Deccan Traps** from India, presently **covering most of the Maharashtra plateau**, are a much larger flood basalt province.
- It is believed that initially the trap formations covered a much larger area than the present.

EXTRUSIVE LANDFORMS

4 MID-OCEAN RIDGE VOLCANOES

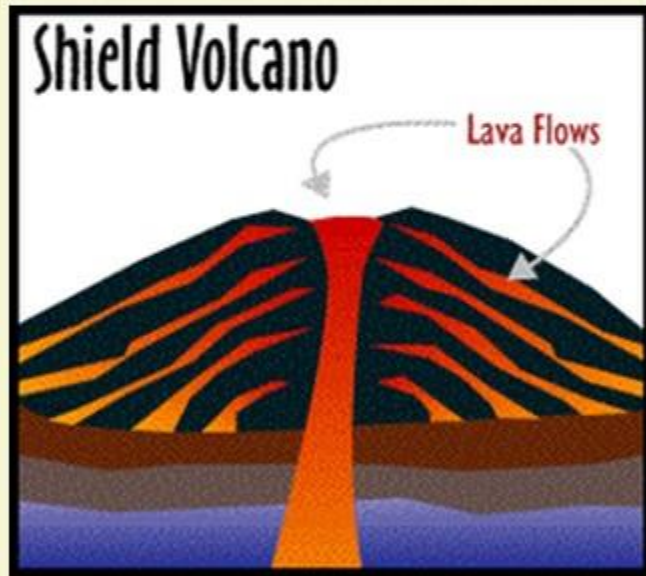


- These volcanoes occur in the oceanic areas.
- There is a system of mid-ocean ridges more than 70,000 km long that stretches through all the ocean basins.
- The central portion of this ridge experiences frequent eruptions.

EXTRUSIVE LANDFORMS

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SHIELD VOLCANOES



1 A shield volcano is a **very large, broad, and low profile volcano** consisting of layers of basaltic rocks. The **Hawaiian Volcanoes** are the most famous examples.

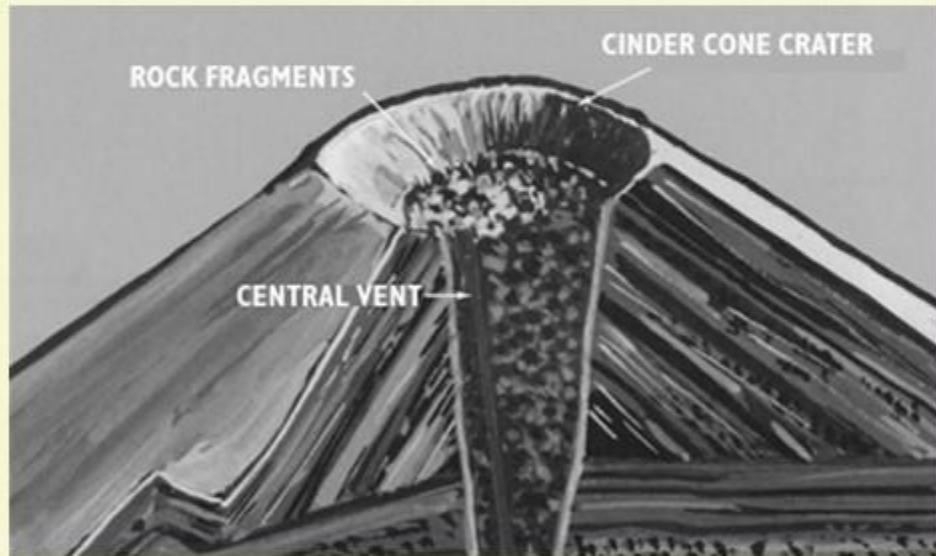
2 Shield volcanoes most **commonly formed in the middle of oceanic plates or in continental rifts**, which are areas in which the crust is being pulled apart.

3 The shape of shield **volcanoes resembles the round shields used by warriors of ancient times to protect themselves in battle.**

EXTRUSIVE LANDFORMS

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CINDER CONES

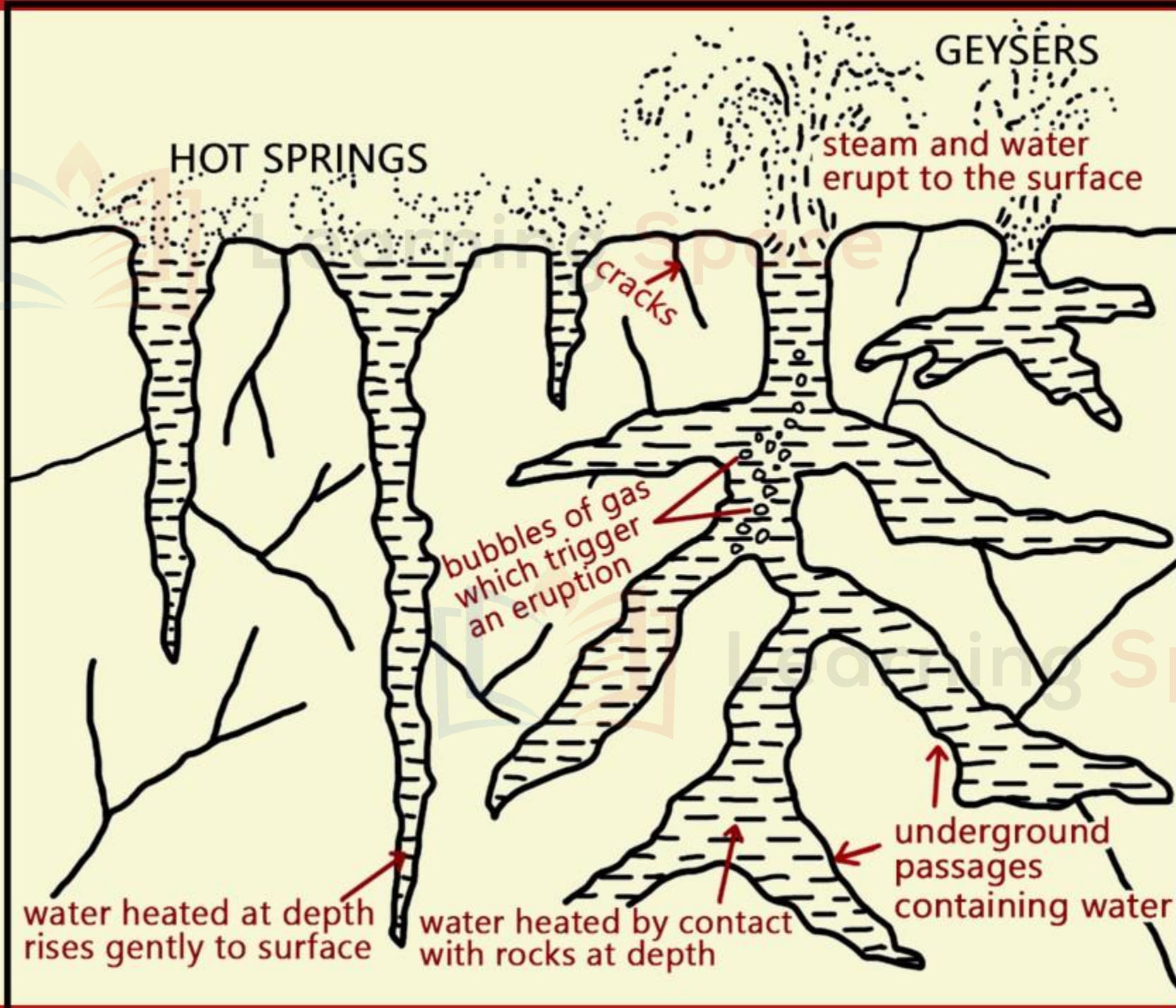


- 1 A cinder cone is a **steep conical hill of loose pyroclastic fragments**, such as either volcanic clinkers, cinders, volcanic ash that has been built around a volcanic vent.
- 2 As the **gas-charged lava is blown violently** into the air, it breaks into small fragments that solidify and fall as **either cinders, clinkers, or scoria around the vent** to form a cone that often is symmetrical.
- 3 They consist of loose pyroclastic debris formed by explosive eruptions or lava fountains from a single, typically cylindrical vent.

WHAT DO YOU UNDERSTAND BY GEYSERS?

- 1** Geysers are **fountains of hot water and superheated steam** that may spout up to a height of 150 feet from the Earth beneath.
- 2** The phenomenon is associated with a **thermal or volcanic region** in which the water below is heated beyond boiling point of 100°C .
- 3** The jet of water is usually emitted with an explosion and is often **triggered off by gases seeping out of the heated rocks**.
- 4** Almost all the world's geysers are confined to three major areas i.e., Iceland, the Rotorua district of North Island, New Zealand and Yellowstone Park of USA.
- 5** World's best known geyser is perhaps the "**Old Faithful**" in Yellowstone National Park, which erupts at regular intervals, every 63 minutes on an average.

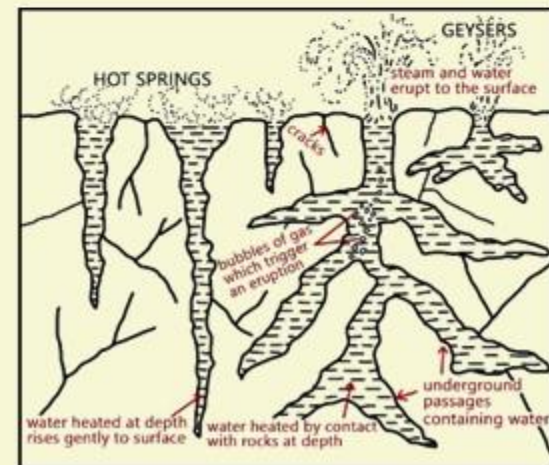






WHAT DO YOU UNDERSTAND BY HOT SPRINGS?

- 1 Hot Springs are also known as **Thermal Springs**.
- 2 They **are found in any part of the Earth**, where water sinks deep enough beneath the surface to be heated by the interior forces.
- 3 The **water rises to the surface without any explosion**. Such springs **contain dissolved minerals** which may be of some medical value.
- 4 Iceland has thousands of hot springs. Some of them have been harnessed to heat the houses, swimming pools and for other domestic purposes.
- 5 Hot springs and Geysers have become tourist attractions in Japan and Hawaii Islands of USA.



GEYSERS AND HOT SPRINGS



GEYSER



HOT SPRING

*A geyser **THROWS UP JETS** of hot water alternating with steam at regular intervals while springs are sources of water from which the **WATER FLOWS OUT** on its own.*





CHINA

North
Korea

South
Korea

Sea of Japan

Tsushima Strait

North Pacific Ocean

JAPAN

East China Sea

North
Pacific
Ocean

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